

Frequency distribution tables

By size and month

Men shoes sales

US	United States, 2016												Mean	Standard error	ME	95% CI	Number of pairs
	1	2	3	4	5	6	7	8	9	10	11	12	2016	2016	2016	2016	
6	4	1	3	1	3	3	3	4	3	7	3	0	2.92	0.51	1.13	1.78 4.05	4
6.5	3	2	0	1	0	0	1	7	2	1	2	1	1.67	0.56	1.22	0.45 2.89	3
7	0	0	1	0	6	4	4	2	3	0	0	0	1.67	0.61	1.34	0.33 3.00	3
7.5	3	2	3	1	7	0	7	3	4	6	1	1	3.17	0.69	1.53	1.64 4.70	5
8	7	9	7	3	12	2	9	4	7	5	2	6	6.08	0.88	1.94	4.14 8.03	8
8.5	12	12	8	8	15	9	17	17	6	9	10	6	10.75	1.12	2.47	8.28 13.22	13
9	17	13	13	11	21	22	25	30	26	25	13	10	18.83	1.97	4.33	14.50 23.17	23
9.5	19	25	27	24	26	33	25	47	31	44	37	26	30.33	2.45	5.39	24.95 35.72	36
10	17	26	26	19	16	31	25	24	23	31	15	20	22.75	1.57	3.45	19.30 26.20	26
10.5	13	16	22	14	28	19	18	15	19	21	16	10	17.58	1.37	3.01	14.57 20.59	21
11	5	16	13	10	10	11	15	8	9	7	6	7	9.75	1.01	2.22	7.53 11.97	12
11.5	4	3	6	3	3	5	6	4	5	12	13	5	5.75	0.96	2.12	3.63 7.87	8
12	3	0	0	4	4	4	3	12	4	9	2	1	3.83	1.01	2.23	1.60 6.06	6
13	1	1	2	0	3	2	1	0	0	4	3	2	1.58	0.38	0.83	0.75 2.42	2
14	2	6	3	3	5	3	2	1	0	1	2	1	2.42	0.50	1.10	1.32 3.52	4
15	0	0	0	1	1	0	4	0	0	0	0	2	0.67	0.36	0.78	-0.12 1.45	1
16	0	0	0	0	0	0	0	0	0	0	0	0	0.00	0.00	0.00	0.00 0.00	0
Total	110	132	134	103	160	148	165	178	142	182	125	98					

$$n = \frac{2016}{12}$$

$$t_{11,0.0} = 2.20$$

The correct t-statistic is 2.20, unlike the 2.18, shown in the video.

This small correction does not change the result in a way, which leads to a different interpretation.

Standard Error

A way to measure how much a “sample statistic” (like an average) is likely to differ from the true value of the entire population.

Formula

$$SE = \frac{\sigma}{\sqrt{n}}$$

SE = standard error of the sample

σ = sample standard deviation

n = number of samples

Margin of Error

is a number that tells you how much “wiggle room” there is in a survey or study. It represents the range above and below a result where the true answer likely sits.

$$\text{Margin of Error} = z \times \frac{\sigma}{\sqrt{n}}$$

Confidence Interval

$$\text{Point Estimate} \pm (\text{Critical Value} \times \text{Standard Error})$$